Jeremiah J. McElwee

Department of Earth System Sciences
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Research Statement

I am a geodynamicist and computational scientist who uses numerical modeling to study fluid-rock interaction and deformation in the lithosphere. I am interested in finding ways to bridge micro-scale and macro-scale processes in the evolution of earth. I believe that geodynamic modeling benefits from multidisciplinary perspectives, and I work to incorporate observations from metamorphic petrology, experimental rock physics, and geophysics into my research. I see a strong benefit in international collaboration and I have proven experience working with international and multidisciplinary groups. My current work focuses on reaction-induced fracturing during hydration reactions and seeks to apply the results to fluid migration, serpentinization, and deformation in the oceanic lithosphere and the mantle wedge corner.

Academic History

Yonsei University Fall 2026 – Present
Postdoctoral Resesarcher
Supervisor: Changyeol Lee

University of Minnesota, Twin Cities Fall 2021 – 2026
PhD, Geodynamics
Advisor: Ikuko Wada

University of California, San Diego Fall 2016 – Spring 2020
B.S., Physics with a minor in Earth Science

Research Experience

Postdoctoral Researcher Fall 2026 – Present
ESS at Yonsei University, under Professor Changyeol Lee

Graduate Research Assistant Fall 2021 – Fall 2026
EES at University of Minnesota, under Professor Ikuko Wada

Undergraduate Research Assistant Summer 2019
IGPP at Scripps Institute of Oceanography, under Professor Dave Stegman

Undergraduate Research Assistant Summer 2018
SERF at University of California, San Diego, under Doctor Carl Melis

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Teaching Experience

Earth and Environmental Sciences Web Accessibility Coordinator Fall 2025

Lab Instructor
ESCI 2201 (Solid Earth Course) Fall 2021 – Fall 2025
ESCI 1003 (Introductory Paleontology Course) Spring 2022

Private Algebra and Geometry Tutor Fall 2020 – Summer 2021

Peer Reviewed Publications

McElwee, J.J., Wada, I., Yoshida, K., Shimizu, H., & Okamoto, A. (2026). Impact of differential stress on fracture due to volume increasing hydration. *Solid Earth, 17(3)*, 537–554. https://doi.org/10.5194/se-17-537-2026

Preprints and Manuscripts in Review or in Prep

McElwee, J.J., Wada, I., Yoshida, K., Okamoto, A., Effects of reaction weakening and reaction-induced stresses on fracture. In review at *JGR: Solid Earth*
https://doi.org/10.22541/essoar.15004644/v1 [Preprint]

McElwee, J.J., Wada, I., Yoshida, K., Okamoto, A., Impact Reaction-Induced and Tectonic Stresses on Fracture and Serpentinization in the Mantle Wedge: Role of Effective Confining Pressure. Submitted to *JGR: Solid Earth* Summer 2026

Presentations and Workshops

"Effects of reaction weakening on reaction-induced fracturing: insights from numerical simulations"
Geo-processes in Mineral Carbon Storage Workshop 2026 Poster

"Impact of Reaction-Driven Fracturing on Fluid Flow and Hydration in the Lithosphere"
UMN Earth Student Research Symposium 2026 Oral

H43E "Localization of reaction-induced fracture and hydration due to reaction softening: insights from numerical simulations"
AGU Fall 2025 Poster

T43E "Effect of Confining Pressure on Reaction-Induced Fracture in the Mantle Wedge Corner"
AGU Fall 2025 Poster

"Reaction-induced fracture during hydration of the lithosphere: insights from numerical simulations"
Georgia Tech Solid Earth Seminar Fall 2025 Invited Oral

"Effects of Reaction Softening and Reaction-Induced Stresses on Fracture in the Mantle Wedge Corner"
CIG: The Next Generation Workshop Summer 2025 Poster

"Effects of Reaction Softening and Reaction-Induced Stresses on Fracture at the Base of the Mantle Wedge Corner"
Fluids in Cascadia CRESCENT Topical Workshop Spring 2025 Poster

T53C-3247 "Effects of Reaction Softening on Reaction-Induced Fracture during Serpentinization of the Forearc Mantle Wedge Corner"
AGU Fall 2024 Poster

"Tectonic and reaction-induced stresses control fracture, fluid flow, and serpentinization in the mantle wedge corner"
Montreal Fluids Workshop Spring 2024 Poster

T52B-04 "Serpentinization in the Forearc Mantle Wedge: Effects of Tectonic and Reaction-Induced Stresses on Fracture Propagation and Fluid Flow"
AGU Fall 2023 Oral

OC2-12 "Mantle wedge serpentinization in subduction zones: Effects of reaction-induced and tectonic stresses on fracture propagation and fluid flow"
WRI17/AIG14 Summer 2023 Oral

T32E-0226 "Propagation of Serpentinization in the Mantle Wedge Corner"
AGU Fall 2022 Poster

Attendee
SEG-AGU Geophysics of Convergent Margins Summer 2022 Attendee

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Awards and Fellowships

Earth and Environmental Sciences Outstanding Teaching Assistant 2024 – 2025

Earth and Environmental Sciences Harold Mooney Fellowship and Banerjee Fellowship 2024 - 2025

Earth and Environmental Sciences Richard Clarence Dennis Graduate Fellowship 2022 – 2023

Service

General Elementary School, Middle School, and Museum Education Activities
Intermittent

Earth and Environmental Sciences Confronting Colonization Working Group
Fall 2024 – Summer 2025

Earth and Environmental Sciences Incoming Graduate Student Mentor
Fall 2023 – Present

Earth and Environment Display Committee
Fall 2022 - Summer 2025

Graduate Student Representative to Earth and Environmental Sciences Faculty Meeting
Fall 2023 - Summer 2024

Earth and Environmental Sciences Outreach Committee
Fall 2021 - Summer 2023

Earth Science Activities for Summer Camps
Summer 2022

References

Ikuko Wada: iwada@umn.edu

Lars Hansen: lnhansen@umn.edu

Atsushi Okamoto: atsushi.okamoto.d4@tohoku.ac.jp